

Part C – System Planning and Design:

1. System Architecture – System Structure:

ERT is a solo-operated online store; it was founded by just one person (the founder) but has a team of digital agents aiding various tasks. The essence of it is that the founder does not do all themselves but divides out tasks which are delegated to agents, where each agent covers some subset of the work.

If the founder must do it (track an order in the system, they use it to create or update a product within. Her task is then sent to the right agent according to the type. The same task is performed by each of the agents in their own manner, giving a result/output based on their skills. And then the founder looks at these and decides. This approach retains operations in order and simplifies store management Because it relies on one single entity.

2. Multi-Agent Structure:

- **Marketing Manager:**

Responsible for the store's marketing; that is, they propose ways to attract customers to the store or manage promotional campaigns. Some proposals may include identifying in-demand products and ideas for increasing sales.

- **Order Agent:**

Responsible for orders; they follow them from start to finish and have a process to ensure everything is organized and error-free.

- **Customer Support Agent:**

Also handles customer service-related tasks, responds to inquiries, and assists customers when they encounter any issues. This improves customer experience.

- **Product Agent:**

Manages products within the store, which involves either adding new products or updating their information to keep it current.

3. System Classes:

Company Category:

This category describes the store as a system with all products and all agents connected to it.

Owner Category:

This is the owner of the store; he creates tasks and assigns agents and then he can make decisions based on reports generated by agents.

Agent type:

This section consists of all agents in the system, where each agent will have a specific Work role.

Task Category:

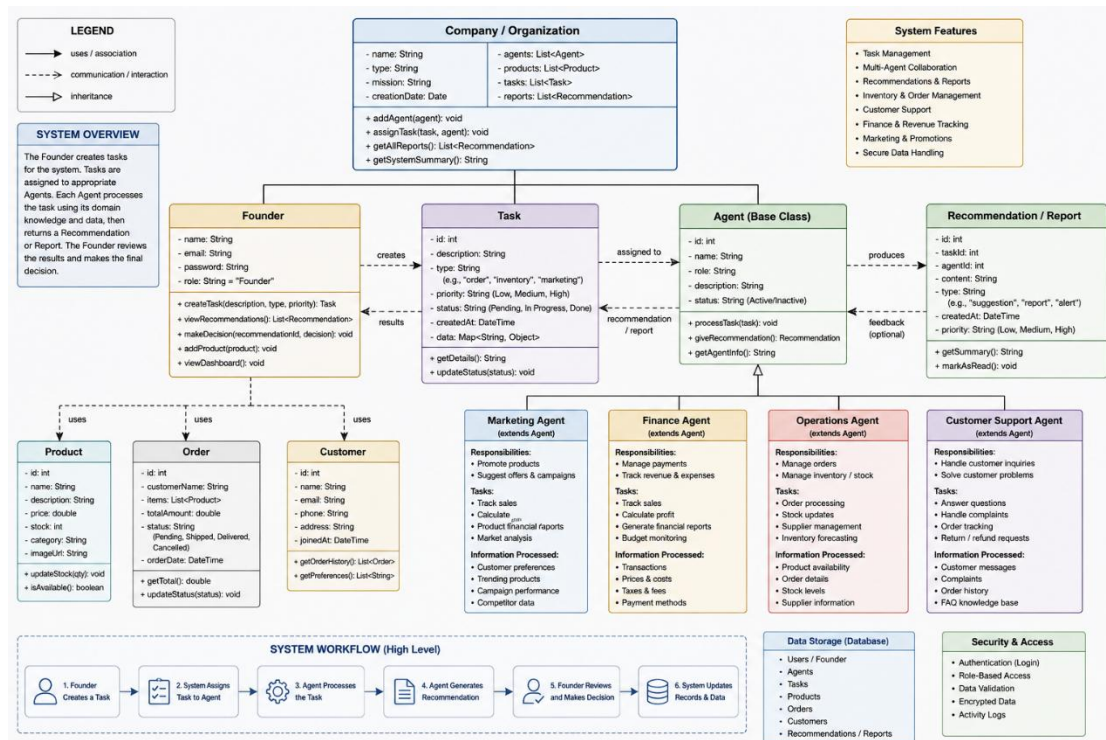
This category represents the tasks which are to be done in the system. Each task will have an ID and description.

Recommendation Category:

This is to say the service or advice the agent gives you after being done with work.

Product Category:

The details about the products which include their names



1. Russell, S. J., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach*. Pearson.
2. IBM. (2024). *What are Intelligent Agents?* <https://www.ibm.com>
3. Python Software Foundation. (2024). *Python Documentation*. <https://docs.python.org>

Google Colab:

<https://colab.research.google.com/gist/hduyu22/7c29c876911754361945d082f2c808ee/untitled1.ipynb>

GitHub:

<https://github.com/hduyu22/Designing-a-One-Person-Company-Using-a-Multi-Agent-System.git>